



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1017 – SOFTWARE ENGINEERING

INDIVIDUAL REFLECTION REPORT

INTELLIGENT DISASTER EARLY WARNING AND EVACUATION MANAGEMENT SYSTEM

NAME: RAKKESH KARAN R

REGISTER NO.: 192524253

ROLE: Frontend Architecture and Application Interface

DESIGN / DEVELOPMENT DECISION: I worked on the frontend structure, navigation, dashboard layout, reusable components, and responsive interface. I selected a persistent navigation structure with clear severity colours, status badges, KPI cards, tables, and action panels because emergency users must locate important information quickly.

CHALLENGE FACED AND SOLUTION: The main challenge was presenting a large amount of operational information without making the interface confusing. I managed this by creating a visual hierarchy, grouping related actions, using consistent components, and planning separate layouts for desktop authority users and mobile citizen users.

LEARNING GAINED: I learned how frontend architecture, accessibility, responsive design, and human-computer interaction affect the usability of a safety-oriented application. I also learned that a visually attractive interface is not sufficient; labels must be clear, important states must be visible, and users must understand what action is expected from them.

SDG RELEVANCE: The project relates to SDG 11 by improving the accessibility and clarity of information needed by communities during hazardous conditions.

COURSE OUTCOME CONTRIBUTION: My contribution supports the course outcome related to web application development and user-centred interface design through the navigation shell, responsive layouts, reusable components, and dashboard screens.

POSSIBLE IMPROVEMENT: I would improve the interface through automated accessibility testing, multilingual support, user testing, and stronger low-bandwidth and mobile support.